THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2012 Fundamental Concepts of Mathematics
(2025-26 Second Semester)

Course Information

Instructor

Dr. Yat-Ming CHAN

Office: Room 312, Run Run Shaw Building

Consultation Hours: TBA

Email: `ymchan@maths.hku.hk`

Phone: 22415198

Tutors

Jingyuan HU

Office: Rm 211B, Run Run Shaw Building

Consultation Hours: TBA

Email: `abbhl126@gmail.com`

Weizhen KONG

Office: Rm 213, Run Run Shaw Building

Consultation Hours: TBA

Email: `kweizhen2022@163.com`

Haoqian LI

Office: Rm 217, Run Run Shaw Building

Consultation Hours: TBA

Email: `u3011713@connect.hku.hk`

Shengbo MA

Office: Rm 217, Run Run Shaw Building

Consultation Hours: TBA

Email: `u3011677@connect.hku.hk`

Dr. Felix SCHREMMER

Office: P206, Graduate House

Consultation Hours: TBA

Email: `schremmer@hku.hk`

Dr. Chiu Yin TSANG

Office: Rm 404, Run Run Shaw Building

Consultation Hours: TBA

Email: `h0347529@connect.hku.hk`

Han WU

Office: Rm 320B, Run Run Shaw Building

Consultation Hours: TBA

Email: `u3009967@connect.hku.hk`

Hanju WU

Office: Rm 216, Run Run Shaw Building

Consultation Hours: TBA

Email: `wuhanju@connect.hku.hk`

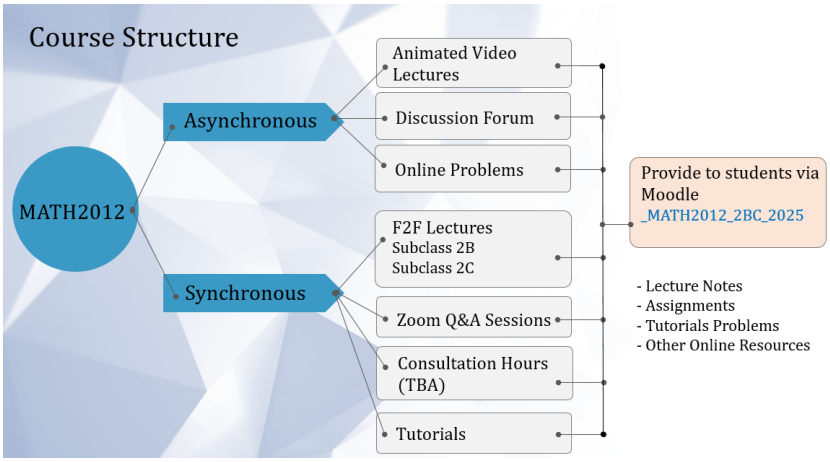
Course websites

Moodle _MATH2012_2BC_2025

Assessment

(1) Final Exam: 50%
(2) Tests: ?% (Test 1: ?% + Test 2: ?%)
(3) Assignments: ?%
(4) Tutorials: ?%

Course Structure



Lectures

Face-to-face (f2f) lectures will be held on
Subclass 2B:
Mondays from **1600 to 1750** at Library Extension Building (LE1) and
Wednesdays from **1100 to 1150** at Library Extension Building (LE1).
Subclass 2C:
Mondays from **1300 to 1450** at Chong Yuet Ming Physics Building (CYPP4) and
Fridays from **1300 to 1350** at Chong Yuet Ming Physics Building (CYPP4).
Note that there will be NO classes on Feb 16, in the Chinese New Year break, in the Reading Week (Mar 9 – 13), and on Mar 16, Apr 3, Apr 6 and May 1.

You are strongly recommended to watch the animated video lectures before joining the f2f lectures. The class time will then be used for the following activities:
(i) review/summary session;
(ii) follow-up and further discussion on materials in video lectures/lecture notes;
(iii) problem-solving and exploration activities
The exact schedule concerning the activities for each week will be announced in moodle.

Animated Video Lectures

Asynchronous learning through animated video lectures gives students flexibility and allows students studying at any time and at their own paces. Be prepared that you will not be able to understand everything instantly while watching the videos. You are expected to spend a substantial amount of time learning the material through your own effort and your interaction with the teacher/tutors/your classmates.

Tutorials

To facilitate the understanding of the course content, there will be 10 (or 11) tutorials scheduled for this term. The **first tutorial** will begin in the week **Feb 2 – 6**. Please go to moodle and choose "*Tutorial Registration*" to indicate your preferable timeslot. The deadline for registration is Jan 30 (Fri), 5:00 pm. You will then be arranged in a tutorial group according to your preference.

Zoom Q&A Sessions

Extra Zoom Q&A and Review Sessions leading up to the submission of assignments or the tests will be arranged. You can ask any questions arising from video lectures/assignments/tutorials. Meeting details will be announced in due course.

Tests

There will be 2 class tests. The first will take place on **Mar 7 (Sat)** during the usual class time, while the second will take place near the end of the semester (probably on **Apr 25 (Sat)**).

Course Description

This course is designed to provide students with solid background on fundamental concepts of mathematics and methods of mathematical proofs. Such concepts and methods are important for subsequent studies in all higher level courses in mathematics. This course can be taken concurrently with other Level 2 or above courses.

Outline of Topics

1. Sets and Logic

   Basic concepts in the language of sets, Subsets, Venn diagrams, Power sets, Set operations, Statements, Conjunctions, Disjunctions and Negations, Conditionals, Tautologies and Contradictions, Quantifiers, Quantified Statements

2. Proof Strategies

   Direct Proof, Proof by Contrapositive, Proof by Cases, Proof by Contradiction, Proof by Mathematical Induction, Counterexamples

3. Equivalence Relations and Functions

   Relations, Reflexive, Symmetric and Transitive Relations, Equivalence Relations and Equivalence Classes, Partitions, Functions as Relations, Injective, Surjective and Bijective Functions, Compositions, Inverse Functions, Denumerable, Countable and Uncountable sets, Comparing Cardinalities of Sets

4. Other Topics (optional)

   Introduction to Calculus: Sequences and Series; Introduction to Group Theory

Reference Books

Main Reference

1. Class lecture notes — This course will follow closely with the lecture notes which can be downloaded from moodle.
2. G. Chartrand, A.D. Polimeni and P. Zhang, *Mathematical Proofs: A Transition to Advanced Mathematics*, Pearson/Addison-Wesley
3. R. Courant and H. Robbins, *What is Mathematics? An Elementary Approach to Ideas and Methods*, Oxford University Press, 2nd edition, revised by I. Stewart

Suggested Readings

1. P.R. Halmos, *Naive Set Theory*, Springer-Verlag
2. K. Houston, *How to Think Like a Mathematician: A Companion to Undergraduate Mathematics*, Cambridge University Press
3. K.T. Leung and D.L.C. Chen, *Elementary Set Theory*, Hong Kong University Press
4. K.T. Leung, P.H. Cheung, *Fundamental Concepts of Mathematics*, Hong Kong University Press
5. C. Schumacher, *Chapter Zero: Fundamental Notions of Abstract Mathematics*, 2nd Edition, Addison-Wesley
6. I. Stewart, *Concepts of Modern Mathematics*, Dover Publications